

Organizational memory influences new product success

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Abstract

Prior research on the impact of organizational memory on new product success has divergent perspectives. Such inconsistency has accrued mainly from not considering memory's multifaceted aspects, which interact with specific project characteristics. This paper tries to sort out this paradox by proposing that project innovativeness moderates the relationship among variables of organizational memory and new product success. An empirical study of 169 NPD projects of Korean manufacturing firms finds that memory sharing and the use of external information and formal procedures enhance new product success, whereas organizational memory has no effect. Project innovativeness is found to moderate memory's effect, despite some tradeoffs. As innovativeness increases, the effect of organizational memory and use of external information become stronger whereas the effect of memory sharing and use of formal procedures weaken. This suggests that firms must activate organizational memory more carefully according to project characteristics and the memory level to maximize its positive effects.

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1. Introduction

New product development (NPD) is one of the key driving forces for attaining competitive advantage and sustained growth of the firm. Because of this, many scholars and practitioners struggle to find the factors that lead to NPD success. The importance of NPD notwithstanding, it seems there are no explicit practical guidelines for successful product development. Part of the answer lies in the fact that the NPD process includes a complex interaction with organizational factors and external environmental factors. That is, the process itself entails great uncertainty (Kanter, 1988; Van de Ven, 1986).

In this respect, many organizational learning theorists view NPD as a learning process because its problem-solving aspects turn individual team members' knowledge into collective team knowledge (Argyris and Schön, 1978; Imai et al., 1985; Sinkula, 1994). Some scholars contend that a learning capacity that surpasses that of competitors is the only source of a firm's

competitive advantage (De Geus, 1988; Dickson, 1992; Slater and Narver, 1995). This leads to the recognition that organizational learning and information processing abilities are the crucial sources of NPD success (Leonard-Barton, 1992; Lynn et al., 2000; Madhavan and Grover, 1998). Huber (1991) argues that organizational learning processes follow four distinguishable phases: knowledge acquisition, information distribution, information interpretation, and organizational memory. Researchers focus mainly on the first three stages, whereas relatively little attention is paid to organizational memory as the storage of knowledge that results from learning.

Scholarly research results also differ on the impact of organizational memory on performance. Researchers who champion the positive effects insist that organizational memory should enhance performance because a firm's long-standing experience promotes efficiency (Cyert and March, 1963; Duncan and Weiss, 1979). Other researchers point to memory's deleterious effects, observing that change from prior patterns becomes more difficult when memory in a particular domain increases (Dougherty, 1992; Leonard-Barton, 1992). They indicate that the effect is not straightforward, but has a more complex and contingent role in product development. Some also argue that the way memory is manifested and shared in organizations is more

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important than its absolute amount (Brockman and Morgan, 2003; Moorman and Miner, 1997).

In this study, we extend previous research in two ways. First, we look at the effect of the various facets of organizational memory: amount of memory, memory sharing, use of external information, and use of formal procedures. Past research incorporating memory sharing as a component of organizational memory stems from the perspective of looking at memory sharing as project team members' collective interpretation of existing organizational memory. In contrast, a significant distinction of this study is that we look at memory sharing of NPD project team members independently of organizational memory. This is because, in innovative projects, many activities can generate new learning apart from existing organizational knowledge.

Second, we propose that memory's impact depends on the degree of project innovativeness. In NPD research, project characteristics are not considered in general. However, the specific type of projects requires different approaches and management styles. Therefore, general conclusions that do not consider project innovativeness may have limiting practical implications (Dewar and Dutton, 1986).

The major contribution of this paper is that we sort out the specific conditions under which organizational memory exerts a positive influence on NPD success and thus we shed some light on the ambivalent relationship between memory and NPD performance. Our results indicate some interesting interaction patterns between project innovativeness and memory-related variables. In projects that require innovativeness, organizational memory and external information help improve NPD performance. In contrast, memory sharing and the use of formal procedures have a negative impact on NPD success. These two findings imply that even though organizations need to build their knowledge base with innovations, it is even more important that they manage their learning so that it can actually be transformed into more productive outputs.

This paper is organized as follows. First, it briefly reviews previous research in organizational memory and NPD. Then it proposes eight hypotheses, four of which address the main effects and four the interaction effects. We test these hypotheses with data from 169 NPD projects conducted by Korean manufacturing firms. Finally, based on the empirical results, we derive managerial implications and suggestions for future research.

2. Prior research

The existence of organizational memory presumes that firms must be viewed as mental entities capable of thought (Sandelands and Stablein, 1987; Weick, 1979). The collective activities of organizations that process, use, and store information can be seen as being distinct from individual activities (Cyert and March, 1963; Daft and Weick, 1984; Huber, 1991), and the resulting stored information constitutes organizational memory (Duncan and Weiss, 1979; Moorman and Miner, 1997). For example, standard operating procedures can drive behavior even when people within the organization no longer individually remember the experiences that generated the routines. This clearly indicates the existence of an organizational memory that is different from the sum of individual memories.

According to Moorman and Miner (1997), organizational memory manifests itself in three basic forms. First, it is found in cognitive elements such as organizational beliefs, knowledge, frames of reference, models, values, and norms. Second, the organization's experiences are encoded in behavioral elements such as formal and informal behavioral routines, procedures, and scripts. Third, physical artifacts that include organizational structure and physical layout also embody memory. Walsh and Ungson (1991) argue that organizational memory is stored in the firm's individuals, culture, transformation, structure, ecology, and external archives. In this way, it is dispersed in various storage bins, not stored in a central storage system like a computer's memory. This makes it difficult to encode, store, and retrieve. It is also hard to change, because memory is accumulated and reinforced over a long period of time (Levitt and March, 1988). Thus, organizational memory acts as a filter in organizing and categorizing incoming external information (Hedberg, 1981).

Moorman and Miner (1997: p. 93) define organizational memory as "collective beliefs, behavioral routines, or physical artifacts that vary in their content, level, dispersion, and accessibility." Basically, collective knowledge gained through experience is stored in various forms in the organization.

Previous research on the effects of organizational memory shows conflicting perspectives. Cohen and Levinthal (1990) argue that organizational memory, as reflected in prior R&D investments, can enhance a firm's ability to assess and import new outside information, which could promote performance. They call this ability "absorptive capacity" and suggest that "fortune favors the prepared firm" (Cohen and Levinthal, 1994: p. 237). Day (1994) views capabilities as complex bundles of skills and collective learning, exercised through organizational processes. Much NPD research also stresses the importance of knowledge and experience as key antecedents of NPD success. For example, through their meta-analysis of NPD research, Montoya-Weiss and Calantone (1994: p. 407) find that successful firms "develop a superior product that is attuned to customer wants and needs, and they have strong marketing and technical knowledge and skills to develop and launch the product." Moorman and Miner (1997) empirically show that organizational memory has a positive impact on short-term financial performance.

However, scholars with negative perspectives point to memory's inflexibility. Specifically, they argue that if organizational memory is embedded in the form of routines, flexibility is then hampered. This effect has been referred to as a "competency trap" (Levitt and March, 1988), "core rigidity" (Leonard-Barton, 1992), and "routine rigidity" (Dickson, 1992). In the NPD area, qualitative studies also report that higher levels of memory have a negative impact on performance because it inhibits any actions outside preexisting action patterns (Dougherty, 1992; Ghemawat, 1991; McDonough, 1993), especially in developing innovative products. Nevertheless, much NPD research shows that routines also enhance success because they reduce the chances of poor outcomes by decreasing the number of costly errors.

We contend this inconsistency lies in the fact that each researcher views memory's specific facets and does not consider project type. Those who argue the positive effects of memory stress the cognitive aspects of it, especially fundamental knowledge such

as R&D investment, and focus more on radical projects. However, those on the opposite side focus on the procedural aspects, such as behavioral routines in more incremental project settings.

In light of these differences, we investigate the effects of various aspects of memory on new product success and seek to explore the moderating role of project innovativeness. We consider the procedural aspects to see if the use of formal processes actually enhances or inhibits new product success. We also investigate the effects of external information and memory sharing. NPD teams must continuously acquire new outside technical and market knowledge because memory generally does not exist in exactly required forms and content. Hence, it may be more important to have the ability of combining various knowledge bases than have memory per se. In this respect, effective and efficient memory-sharing systems through open communication and team cohesiveness play a critical role in new product success. In addition to the main effects of these four variables, we explore the moderating effects of project innovativeness on each memory-related variable and the new product success relationships. Fig. 1 depicts the research model.

3. Research model

3.1. Main effects

The organizational memory accumulated through a firm's experiences constitutes a valuable knowledge asset and acts as a source of competitive advantage (Prahalad and Hamel, 1990; Winter, 1987). In an NPD setting, many researchers show that NPD teams with expertise in product-related technology and markets perform better (Brown and Eisenhardt, 1995; Henard and Szymanski, 2001). This is because memory increases efficiencies and decreases possible repeated errors (Cyert and March, 1963; Duncan and Weiss, 1979). Much NPD research confirms this positive effect on new product success (Cooper and Kleinschmidt, 1986; Moorman and Miner, 1998). Moorman and Miner (1997) empirically validate the belief that memory enhances short-term financial performance.

As mentioned before, some previous research points to memory's negative effects on new product success. However, we argue that memory will have a positive effect because of its potential to extract marketing and technological synergy between the new product and the firm's existing competencies (Montoya-Weiss and Calantone, 1994; Zirger and Maidique, 1990). Therefore, we hypothesize:

H1. Organizational memory is positively related to the new product success.

Sharing a memory with organizational members means that teams can accomplish a job in a time-effective way through open communication. Because the required knowledge for NPD is dispersed through the firm's functional areas, sharing understanding and integrating information between the functional groups become key drivers for success. The literature on information-sharing mechanisms in cross-functional efforts suggests that knowledge and vision sharing improve the

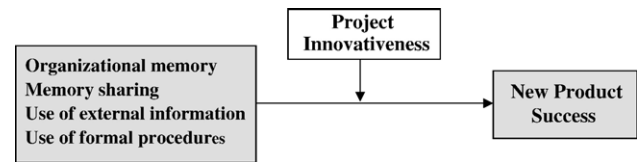


Fig. 1. Research model.

financial performance of product development activities by enhancing cross-functional understanding and cooperation (Day, 1994; Imai et al., 1985). Moreover, knowledge sharing helps promote a common interpretation of incoming information, which is an important condition for organizational learning (Day, 1994; Slater and Narver, 1995). In a similar vein, much research shows that the use of multifunctional teams has a positive effect on new product innovation (Gupta et al., 1986; Moenaert and Souder, 1990a,b). This stream of research suggests that memory sharing enables functions to understand one another, improves their ability to cooperate, facilitates cross-fertilization, and may reduce the tendency of individual functions to become confined by their own thought-worlds (Moorman and Miner, 1997). Moorman and Miner (1997) show the dispersion of organizational memory is positively related to a firm's performance. Therefore, we hypothesize that the greater the extent of memory sharing, the greater the new product performance. Thus:

H2. Memory sharing is positively related to new product success.

NPD teams must acquire outside information. This is because, in general, organizational memory does not contain all the information needed to develop a new product. From a practical standpoint, firms tend to develop a product by combining already known external technologies while not performing basic R&D because of time constraints. In this respect, March and Simon (1968) argue that most innovations are borrowed from external ideas. In addition, von Hippel (1988) suggests that innovation originates from outside sources, especially customers, because a successful product requires careful investigation of customer preferences and adaptability as well as technical specifications. This view is supported by many marketing scholars (Cooper and Kleinschmidt, 1995; Joshi and Sharma, 2004; Li and Calantone, 1998).

Baker and Sinkula (1999) suggest that a new product's performance is enhanced if the synergy between an inside-out learning orientation and an outside-in market orientation exists. In other words, the effective use of external information is known to affect new product success by exploiting the synergy between internal competencies and external knowledge (Brown and Eisenhardt, 1995). Thus:

H3. Use of external information is positively related to new product success.

Organizational memory is stored in various retention bins in an organization (Walsh and Ungson, 1991), so an effective

memory retrieval system is essential. To this end, firms generally generate a routine that contains experience and knowledge (Cyert and March, 1963; Garud and Nayyar, 1994). In fact, Intel effectively performs NPD by using its resources wisely, such as reusing proven parts, using statistical quality control procedures to reduce the production of defective parts, and designing for manufacturability (Gupta and Wilemon, 1990). Many researchers also note that explicit development objectives coupled with the implementation of systematic development processes is one of the most robust antecedents of new product success (Griffin, 1997; Kessler and Chakrabarti, 1996; Lynn et al., 1999). Dröge et al. (2000) assert that firms tend to adopt environmental uncertainty by routinizing production technology, and those with well-formed routines show good performance by efficiently using prior knowledge. Hence, the use of formal procedures, mostly manifested in various routines, should positively affect new product performance. Therefore, we hypothesize:

H4. Use of formal procedures is positively related to new product success.

3.2. Interaction effects

As posited earlier, the inconsistent research results on organizational memory and new product development stem from a lack of consideration of specific project types. Prior research has emphasized the importance of investigating success factors according to project characteristics (Dewar and Dutton, 1986; Olson et al., 1995; Shenhar, 2001). Although many typologies exist, and researchers seem to use their own specific variations (Garcia and Calantone, 2002), a widely used dichotomy is to classify projects as “radical” or “incremental” (Utterback, 1996).

Radical projects usually require fundamental technological changes for a firm and must cater to completely new market and customer needs. Incremental projects focus mainly on minor product revisions or product line extensions for existing markets and customers and thus do not accompany breakthrough technology or make much difference in a firm’s stored product-market experience (Chandy and Tellis, 1998). In reality, most projects fall into some intermediate position on the radical/incremental continuum. A related discussion also differentiates between product and process innovation (Abernathy and Townsend, 1975; Abernathy et al., 1978). The former refers to innovations embodied in the product itself, and the latter deals with innovations involved in the overall manufacturing of the product. While innovation at both levels is equally important for organizations, this paper focuses on product innovation.

For more radical projects, the project team must acquire technological and market information from outside and produce new combinations. Memory-rich firms are more likely to perform better because they excel in their absorptive capacity to effectively acquire and use external information (Cohen and Levinthal, 1990). However, firms that do not have enough

memory may have difficulty in determining the importance and adaptability of outside technological and market information. For incremental projects, even if a firm has enough memory for the product category, it is likely that competitors also have comparable memory because the product category in question is based on well-known technology and market conditions (Roussel et al., 1991; Souder and Song, 1997). Hence, the effects of organizational memory will not be so strong.

Sorescu et al. (2003) investigated 255 breakthrough innovations in the pharmaceutical industry and found that firms having greater depth and breadth in their product portfolios achieved higher performance. Nerkar and Roberts (2004) find that a firm’s technological knowledge and market experience positively affect performance for innovative projects but do not find such a relationship for incremental projects. In a similar vein, Shenhar (2001) asserts that as project innovativeness increases a firm must rely more on internal knowledge. Dewar and Dutton (1986), studying the shoe industry, show that knowledge-abundant firms, as reflected in the number of engineering experts, adopt more innovative projects. Therefore, as the innovativeness of projects increases, firms tend to rely more on memory. This leads us to hypothesize:

H5. The greater the project innovativeness, the stronger the positive relationship between organizational memory and new product success.

Well-shared organizational memory enables common understanding and promotes effective communication among team individuals. As a result, memory sharing reduces unnecessary time-consuming activities and enhances efficiency. Many marketing scholars note this “mental model” effect of team members as one of the important drivers of innovation (Day and Nedungadi, 1994; Senge, 1990).

However, to be innovative, a firm must have diverse as well as a large absolute amount of organizational memory. This is because diverse ideas from individuals can provide “requisite variety”, which becomes the source of new combinations. As a matter of fact, too much memory sharing may incur the risk of falling into “groupthink,” which can seriously hamper organizational innovativeness (Janis, 1982). In this respect, Leonard-Barton (1995) introduced the concept of “creative abrasion,” which is the deliberate conflict of ideas. Of course, whether this abrasion of ideas is creative or destructive may depend on the capabilities of individual specialists to sustain a meaningful and synergistic conversation with one another (Madhavan and Grover, 1998). Iansiti (1993) terms such skills as “T-shaped skills” and asserts that managers should have this capability to innovate.

Nemeth (1997) contends that organizations should not use the traditional approach emphasizing “getting along with team members,” especially in the case of innovative products. Previous research shows that the positive effect of memory sharing on performance weakens when project innovativeness is high (Madhavan and Grover, 1998). Sethi et al. (2001) also indicate that teams showing high cohesiveness produce less innovative output. In practice, every team has a moderate

amount of sharing and cohesion as well as a higher risk of having too much familiarity and consolidation. Therefore, we can hypothesize that as project innovativeness increases, the effect of memory sharing on project success will be hampered, that is:

H6. The greater the project innovativeness, the weaker the relationship between organizational memory sharing and new product success.

Effective acquisition of external information is important because it not only helps widen the organization's memory base, it also deepens the memory, providing needed diversity. Many NPD studies confirm the positive performance effect of external information (Brown and Eisenhardt, 1995; Kessler and Chakrabarti, 1996). Moreover, in the case of innovative projects it is likely that firms lack adequate memory to generate ideas and develop products. Thus, an organization's dependence on external memory would be high. Furthermore, because innovative projects are usually marked by high environmental turbulence and change, it is important for firms to continuously catch up with current technological and market trends (Jansiti and West, 1997). That is, in an environment of high turbulence, organizations must have the ability to sort out appropriate information and link it with internal capability (Moorman and Slotegraaf, 1999). Recent trends in benchmarking and R&D alliances represent corporate efforts to effectively acquire external information (Lambe and Spekman, 1997; Wuyts et al., 2004). Therefore, based on the fact that innovative projects are more uncertain, we hypothesize that as project innovativeness increases, the need for external information to ensure NPD success will grow. Thus:

H7. The greater the project innovativeness, the stronger the positive relationship between the use of external information and new product success.

Innovation requires deviation from existing patterns because if firms simply follow previously established procedures and behavioral patterns they will find it difficult to generate creative ideas. Many scholars who argue about organizational memory's negative effects usually make this point. Dougherty (1992) describes a firm's routine as being a barrier to innovation, and Leonard-Barton (1992) and Levitt and March (1988) assert that an organization's routine has a negative effect on NPD performance because it promotes rigidity. In the same light, Amabile (1988: p. 140) states: "If information is stored according to rigid algorithms, creativity is less probable. But if information is stored in wide categories with easy access of association, increased information should only lead to increased creativity." Of course, a rigid outcome stemming from formal procedures is only one facet of organizational memory. Moreover, the negative effect of formal procedures will be most apparent in innovative NPD projects in which a routine's rigidity may outweigh the efficiency effect.

Gersick and Hackman (1990) also note the deleterious effects of habitual routines and suggest that a self-managing

team is one approach to handling this problem. According to them, teams must (a) develop routines that aid in maintaining flexibility and (b) keep the level of deliberate processing within the group high enough to avoid many of the dysfunctional consequences that come from mindless habitual behavior. Therefore, we hypothesize:

H8. The greater the project innovativeness, the weaker the relationship between the use of formal procedures and new product success.

4. The study

4.1. Measurement

We measure all of the constructs in this study based on scales developed in previous research studies. We operationalize organizational memory with four items measuring the amount of knowledge, experience, familiarity, and degree of R&D expenditures to a given product category (Moorman and Miner, 1997). The assumption underlying this approach is that the more a firm has knowledge, experience, familiarity, and R&D spending in a product category, the more memory will be accumulated in various forms within the firm. Memory sharing focuses on the degree of common understanding and interpretation, resulting in the free exchange of ideas and perspectives. It is measured by five items: the degree of shared knowledge and problems, consensus on the importance of information, shared understanding, quick and easy access to information, and open communication among new product participants (Atuahene-Gima, 2003; Brockman and Morgan, 2003). The use of external information is measured by four items: acquisition of external technological and market information, use of external experts, and systematic benchmarking (Kyriakopoulos and de Ruyter, 2004). The use of formal procedures is measured by four items: degree of formality, standardization, documentation, and project milestones. We conceive formal procedure as being the degree of routinization of organizational memory in the NPD process (Atuahene-Gima, 2003).

New product success, the dependent variable, is measured by sales, market share, ROI, and profit relative to targets, in accordance with the scales used by Moorman and Miner (1997). We use these perceived measures of NPD success because of the difficulty of obtaining objective financial data. Project innovativeness is adapted from Sarin and Mahajan's (2001) measures: introducing product-related innovation, technological innovation, new features, and originality compared to competitor's product, originality compared to similar products in an organization. Previous research tends to measure innovativeness based on the simple radical/incremental dichotomy, and then tests for differences on key variables. In this paper, we measure project innovativeness as a continuous variable so that we can test more elaborate models.

Finally, we enter competitive intensity, team size, and organizational formalization into the model as control variables. We enter competitive intensity and team size to control for the amount of memory (Shenhar, 2001; Souder and Song, 1997), and

organizational formalization to control for the structure of information sharing. Competitive intensity is measured by five items: degree of competition, sales promotion, price competition, the number of competitors, and ease of product imitation (Jaworski and Kohli, 1993). Team size is defined as the number of full-time project participants. Organizational formalization, the degree to which a firm emphasizes rules and procedures (Zaltman et al., 1973), is measured by six items: clarity of responsibility, clarity of roles, the degree of explicit job descriptions, degree of autonomy, degree of formality of reporting relationships, and the degree of checking for rule violations (Ayers et al., 1997; Deshpandé and Zaltman, 1982). All the variables except team size are measured using a seven-point Likert scale.

4.2. Sample

Despite the inherent time and cost, we chose to interview key informants directly rather than rely on the conventional mailing method. This is for two reasons. First, we needed to boost the response rate, which is usually very low with organizational informants. In particular, some respondents felt the questionnaire contained delicate proprietary information and therefore resisted cooperating. Second, NPD-related studies usually suffer from a positive response bias, which we wanted to eliminate as much as possible. We tried to control the sample to have adequate variance in terms of memory and success measurement variables. As stated previously, this study focuses on product innovation, so measures related to process innovation, such as cost reduction endeavors, are omitted in the final survey.

The initial structured questionnaire was pilot tested on four projects, carefully chosen so that there was no bias to any particular industry. The industries include automobile, steel, electronics, and electrical projects. We eliminated inappropriate questions and clarified items where necessary. The pilot tests helped clarify the readability of the questionnaire but, given that the questionnaire items were translated from the original English scales, potential translation biases still exist. The survey was administered via personal interviews, so that respondents were able to receive explanations about each item and ask questions in situations where certain terms were unclear. The finalized questionnaires were administered to key informants in major Korean manufacturing firms who had recent experience with participating in NPD projects as project leaders or members. In the end, 169 usable questionnaires were obtained on which we empirically test our model. The final sample consists of NPD projects in the following sectors: automobile (7.6%), electrical (14%), electronic (39.5%), telecommunication (14.5%), steel (15.7%), chemical (5.8%), and others (2.9%). The reason for the relatively large presence of electrical and electronic projects is that these industries have many products and many cases were reported.

4.3. Methodology

Following data collection, measures were subjected to a “purification process” involving unidimensionality, discrimi-

nant validity, and reliability assessments (Bagozzi and Phillips, 1982; Churchill, 1979; Gerbing and Anderson, 1988). The unidimensionality of the model is confirmed when the six-factor model corresponding to the six model variables has adequate fit to the data. However, Bentler and Chou (1988) recommend that the ratio of sample size to parameter estimates must not exceed five-to-one. So we tested the model fit of 15 ($6 \times 5/2$) models consisting of every possible pair of variables. Results suggest that the 15 models fit relatively well, as shown in Table 1.

Within these models, discriminant validity was assessed by comparing two models: one with the phi coefficient fixed to 1 and the other with the phi coefficient free. Constraining the phi coefficient to 1 is to assume perfect correlation, meaning that the two constructs represented in the model are the same. Hence, discriminant validity is verified if the freed model shows better model fit than the constrained model. Table 1 shows that discriminant validity is confirmed for all the models.

Finally, the reliability of the measures exceeds 0.76 and meets the standards for acceptance. Table 2 presents the reliability indices (Cronbach's α) and a correlation matrix of all the measures.

5. Model testing and results

5.1. Model testing

The hypotheses for this study are tested using a linear regression model with new product success as the dependent variable. Independent variables are organizational memory, memory sharing, use of external information, and use of formal procedures. Project innovativeness is used as the moderator

Table 1
Tests of unidimensionality and discriminant validity

Model 1	Unidimensionality				Discriminant validity	
	χ^2 -value	d.f.	CFI ^a	GFI ^b	$\Delta\chi^2$	d.f. ^c
1 Org. memory–memory sharing	71.803	26	0.964	0.919	33.765	1
2 –External information	53.520	19	0.959	0.933	61.224	1
3 –Formal procedures	73.976	19	0.956	0.895	19.428	1
4 –Innovativeness	104.144	26	0.929	0.871	61.489	1
5 –Success	70.115	19	0.957	0.942	41.358	1
6 Memory sharing–external information	105.563	26	0.918	0.883	46.116	1
7 –Formal procedures	97.287	26	0.944	0.888	35.827	1
8 –Innovativeness	148.254	34	0.907	0.844	37.105	1
9 –Success	113.379	26	0.932	0.914	35.524	1
10 Ex. information–formal procedures	81.567	19	0.927	0.889	54.578	1
11 –Innovativeness	115.423	26	0.888	0.862	59.945	1
12 –Success	82.625	19	0.927	0.909	38.073	1
13 Formal procedures–innovativeness	129.487	26	0.907	0.841	50.684	1
14 –Success	112.033	19	0.922	0.909	32.231	1
15 Project innovativeness–success	137.306	26	0.901	0.882	42.779	1

^a CFI: Comparative-fit-index.

^b GFI: Goodness-of-fit index.

^c Critical value of χ^2 (1) is 3.84.

Table 2
Correlation matrix and reliability

	Mean	S.D.	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
(1) Organizational memory	4.66	1.34	<i>0.91</i>								
(2) Memory sharing	5.14	1.07	0.44	<i>0.92</i>							
(3) Use of external information	4.21	1.13	0.26	0.41	<i>0.83</i>						
(4) Use of formal procedures	4.70	1.39	0.60	0.40	0.28	<i>0.93</i>					
(5) Project innovativeness	4.77	0.99	0.25	0.48	0.34	0.28	<i>0.89</i>				
(6) New product success	4.17	1.08	0.33	0.42	0.45	0.38	0.38	<i>0.93</i>			
(7) Team size	11.10	9.35	0.14	−0.07	0.11	0.17	0.14	0.12	–		
(8) Competitive intensity	4.98	1.25	0.22	0.11	0.07	0.25	0.18	0.27	0.13	<i>0.83</i>	
(9) Formalization	5.13	0.87	0.38	0.55	0.26	0.46	0.36	0.27	0.20	0.34	<i>0.76</i>

*The coefficient α for each measure is on the diagonal (and in italics) and the intercorrelations among the measures are on the off-diagonal.

variable and is entered in the model in a multiplicative form with the main independent variables. Because the use of moderator variable inevitably causes multicollinearity, the main effect variables are mean-centered before we constructed the interactions to reduce the potential effect of collinearity (Aiken and West, 1991; Cronbach, 1987). Mean centering has no effect on the significance of the model testing, but makes a slight difference in interpreting the regression coefficients. That is, the regression coefficient of one independent variable is the effect of that variable provided that other variables are at their mean values. We investigate variance inflation factors and find that all are below harmful levels (Mason and Perreault, 1991). In addition to these hypothesized effects, competitive intensity, team size, and organizational formalization also are entered as control variables in the model for reasons cited previously.

5.2. Results

The results show that the regression model is significant, with R^2 equal to 0.372 and $F_{(12, 157)}$ equal to 9.335 (p level equal to .000). All the hypotheses are supported except for the effect of organizational memory (β equal to -0.040 , t value equal to -0.479). But as project innovativeness increases, memory has a significant positive effect on new product success, indicating that project innovativeness shows a strong interaction effect. H2 predicts that well-shared memory would be likely to enhance new product success rate. Results indicate a significant relationship between memory sharing and new product success, which supports H2 (β equal to 0.233, t value equal to 2.551). H3 and H4 predict that the use of external information and use of formal procedures, respectively, will have a positive effect on new product success. Both hypotheses are supported (use of external information: β equal to 0.306, t value equal to 4.371; use of formal procedures: β equal to 0.201, t value equal to 2.346).

The results also support H5, which predicts that the greater the project innovativeness the stronger the effect of organizational memory on new product success (β equal to 0.158, t value equal to 1.979). H6 hypothesizes that as project innovativeness increases, the positive effect of memory sharing on new product success will be weakened; results support this prediction as well (β equal to -0.141 , t value equal to -1.654). H7 posits that a more innovative project will require greater use of external information, and the prediction is supported by the test results

(β equal to 0.163, t value equal to 2.252). And the use of formal procedures in innovative projects has a negative effect on new product success, as hypothesized in H8 (β equal to -0.132 , t value equal to -1.678). Finally, the control variables team size and competitive intensity have positive effects on new product success, but are not found to have a significant relationship with organizational formalization. Table 3 presents the model estimation results.

6. Implications

The general implication of this study is that the relationship between organizational memory and new product success is a complicated one that is significantly affected by project characteristics. As expected, memory sharing is found to be a crucial success factor, which reminds us of the old saying, “It takes more than pearls to make a necklace”. In other words, the mere stocking of memory is not enough if it is not shared, because memory has to be dispersed throughout the organization in the form of individual knowledge, behavioral routines, and organizational structures.

Table 3
Estimation results

Independent variables	Dependent variable (new product success)		
	Hypotheses	Standardized coefficient	t -value
Organizational memory	(H1+)	−0.040	−0.479
Memory sharing	(H2+)	0.233	2.551 **
Use of external information	(H3+)	0.306	4.371 ***
Use of formal procedures	(H4+)	0.201	2.346 **
Innovativeness $\times$ organizational memory	(H5+)	0.158	1.979 **
Innovativeness $\times$ memory sharing	(H6−)	−0.141	−1.654 *
Innovativeness $\times$ use of external information	(H7+)	0.163	2.252 **
Innovativeness $\times$ use of formal procedures	(H8−)	−0.132	−1.678 *
Project innovativeness		0.068	0.911
Team size		0.141	2.067 **
Competitive intensity		0.198	2.966 ***
Organizational formalization		−0.122	−1.461
Adjusted R^2			0.372
F -statistics: $F_{(12, 157)}$			9.335 ***

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The use of external information is important in that firms are not likely to have all the memory required to develop new products, especially in this highly competitive era. Understandably, firms are struggling to absorb external technological, customer, and supplier information. The recent growing popularity of R&D alliances and EVI (early vendor involvement) activities reflect the high salience of these concerns. We also found that formal procedures are generally helpful for firms, which stresses the importance of activities like certification of ISO 9000 and 14000 series as well as Six Sigma.

Unfortunately, these effects are contingent on project characteristics. As project innovativeness increases, more organizational memory and use of external information are required. However, memory sharing and the use of formal procedures can hamper new product performance. These findings suggest that companies must take different approaches in managing project and knowledge structure according to project innovativeness. When developing innovative products, it is important to enable team members to access the stored organizational knowledge. Moreover, systematic scanning of relevant external information must be emphasized. This implies that in innovative projects, the boundary role of the marketing function becomes important.

At a more fundamental level, the tendency of companies to overemphasize short-term incremental projects is worrisome. This is because in today's highly charged competitive environment firms are under tremendous pressure to develop innovative new products faster than other firms (Wind and Mahajan, 1997). Hence, they need the accumulation of organizational memory through R&D. As Cohen and Levinthal (1990) note, absorptive capacity acquired through R&D functions becomes a catalyst in transforming external information into internal knowledge. We agree with Teece (1982), who insists that basic R&D is the ultimate driver of competitive advantage.

However, results of memory sharing and the use of formal procedures indicate that companies must consider the tradeoff effects in organizing and managing project teams. Because project teams generally have a memory-sharing culture (Sethi et al., 2001), firms should find ways to limit the effects of too much team cohesion in order to promote diversity and flexibility. Of course, we do not want to imply that memory sharing is always harmful. Toyota, the world's most successful firm for its continuous improvement (kaizen) production system, operates subtle control mechanisms to secure needed diversity and to maintain a shared culture simultaneously. For example, Toyota intentionally brings problems to the surface even when projects are going as planned. Having team members repeatedly propose and solve problems makes them engage in organizational learning, a practice that has come to be known as "the Toyota Way" (Liker, 2004). In developing the Prius, its first hybrid vehicle, Toyota pushed this strategy further by appointing non-experts as project leaders, thereby starting with a different perspective from the outset.

It is also noted that innovative projects need a more flexible approach rather than strictly following formal procedures. This is because over-dependence on formal procedures may hinder the consideration of better alternatives and innovative ideas. Honda uses a "rugby approach," which involves the entire team

running the full length of the NPD process, in contrast to a "relay approach," which involves functions handing off the product at distinct times (Imai et al., 1985).

7. Future research

This study is meaningful in that the effect of organizational memory in NPD settings, which is rarely found in other studies, is empirically modeled and tested. However, the paper also has some limitations that can form the basis of future research in this area.

First, organizational memory is an abstract construct involving a company's cognitive processes that is very difficult to measure. To make matters worse, the memory resides in diverse storage bins throughout the organization. It can be manifested in the form of knowledge and concepts, or procedures and processes, or organizational structure. Moreover, some individuals in organizations may possess all the memory, or many functions and individuals will each own parts of it. In this study, organizational memory is measured in a rather macro perspective with the assumption that a firm's long-standing history and experience can be transformed into various forms of organizational memory. Further research should focus on measuring the micro-level aspects. For example, Moorman and Miner (1997) suggest four types of organizational memory: level, dispersion, accessibility, and content. Memory also can be categorized into tacit vs. explicit and declarative vs. procedural. Those who argue that the effects of organizational memory are positive in previous research base it on declarative memory, whereas the arguments for negative effects are based on procedural memory. It is therefore imperative that future researchers should distinguish the specific type of memory being studied so that the direction can be predicted properly. Specifically, an intriguing line of inquiry for future research may be to look at the relative effectiveness of declarative or procedural memory in radical versus incremental projects.

Second, it may be useful for further research to study the differential effects of technological and market memory. It could be argued that some projects require more market knowledge than others even though market knowledge is generally important (Joshi and Sharma, 2004; Li and Calantone, 1998). Another interesting topic would be to look at how environmental turbulence affects NPD performance. We suspect that more flexibility is needed in high market turbulence than in high technological turbulence because market change promotes much more uncertainty and unpredictability. Perhaps in situations of high technological turbulence the effect of organizational memory is diminished because of memory obsolescence. In contrast, in situations of high market turbulence the effects become more pronounced for firms that can use their knowledge base to adapt quickly to changing customer needs.

Third, future research is needed to study the speed of NPD as the dependent variable. It is likely that organizational memory per se does not have such a direct relationship. But it can be an important practical contribution if research can find the mechanism through which organizational memory can be linked to the speed of NPD. It will be useful to investigate the effects of information

technology such as expert systems, shared database, and electronic bulletin boards (Huber, 1991; Olivera, 2000).

Finally, even though we believe our sample is well representative of the target population, we still may have some self-selection bias. In NPD research, it is generally hard to solve this problem because informants tend to refer to successful projects. Moreover, it is common for projects to be terminated early in the development phase, with a resulting negative effect on the recollection of the related NPD facts. It is therefore possible that some positive success bias exists despite our efforts to reduce it.

The road to NPD success is long and winding. This study shows that companies can take an important first step toward developing successful new products if they can better activate organizational memory by taking into account project characteristics, memory levels, and system flexibility.

Appendix A. Measures

	Mean	Standard deviation
<i>Independent variables</i>		
<i>Organizational memory</i> (4 items, reliability=0.91)		
Prior to the project, my division had		
–A great deal of knowledge about this product category	4.8	1.49
–A great deal of experience in this product category	4.6	1.58
–A great deal of familiarity in this product category	4.7	1.54
–Invested a great deal of R&D in this product category	4.5	1.45
<i>Memory sharing</i> (5 items, reliability=0.92)		
During the project		
–All the team members shared knowledge accumulated within our firms	5.1	1.21
–All the team members shared similar understanding of the related issues and problems	5.2	1.16
–Project team members engaged in open and honest communication	5.2	1.25
–Access to information from team members was quick and easy	5.1	1.26
–Project team members shared a high degree of consensus in regard to the importance of information	5.2	1.21
<i>Use of external information</i> (4 items, reliability=0.83)		
During the project, project team		
–Regularly discussed with customers	4.3	1.45
–Systematically acquired external technological and market information	4.5	1.32
–Consulted regularly outside experts	3.8	1.42
–Conducted systematic benchmarking	4.2	1.39
<i>Use of formal procedures</i> (4 items, reliability=0.93)		
During the project, the team used		
–Well-defined procedures	4.8	1.51
–A standard approach	4.7	1.55
–Well-defined manuals or documentation	4.6	1.60
–Well-defined project milestones	4.7	1.49
<i>Project innovativeness</i> (5 items, reliability=0.89)		
The product developed by the project		
–Introduced several product-related innovations	5.0	1.06
–Introduced high-quality technical innovations	4.8	1.09
–Introduced completely new features	4.5	1.34
–Offered unique features, attributes, benefits to the customers compared to similar products developed by competitors	4.8	1.21
–Offered unique features, attributes, benefits to the customers compared to similar products developed by our company	4.8	1.28

Dependent variable

New product success (4 items, reliability=0.93)

Rate the extent to which the product has achieved the following outcomes

–Sales relative to goals	4.2	1.17
–Market share relative to goals	4.1	1.21
–ROI relative to goals	4.2	1.15
–Profit margin relative to goals	4.2	1.23

Control variables

Competitive intensity (5 items, reliability=0.83)

Rate the extent of the following items

–Competition in this product category	5.5	1.38
–Frequency of sales promotion in this product category	4.9	1.50
–Price competition in this product category	5.1	1.69
–The ease of product imitation	4.1	1.50
–The number of competitors	4.4	1.54

Organizational formalization (6 items, reliability=0.76)

During the project

–My responsibilities were clearly defined	5.3	1.21
–My role in the team was clearly defined	5.4	1.14
–Team members' job descriptions were explicit	5.3	1.26
–I had considerable decision-making autonomy*	4.5	1.37
–Going through the proper channels in getting this job done was constantly stressed	4.4	1.47
–I was constantly being checked on for rule violations	5.4	1.23

*Reverse coded.

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